

Zero-Shot Sparse Mixture of Low-Rank Experts Construction From Pre-Trained Foundation Models

Anke Tang[✉], *Graduate Student Member, IEEE*, Li Shen[✉], *Member, IEEE*, Yong Luo[✉], Shuai Xie, Han Hu[✉], *Member, IEEE*, Lefei Zhang[✉], *Senior Member, IEEE*, Bo Du[✉], and Dacheng Tao, *Fellow, IEEE*

Abstract—Deep model training on extensive datasets is increasingly cost-prohibitive, prompting adoption of deep model fusion to leverage knowledge from pre-existing models. From weight averaging to more sophisticated methods, fusion effectively improves model performance and accelerates new model development. However, parameter interference between models and the lack of interpretability remain challenges. Existing methods address interference by evaluating parameters attributes, such as magnitude or sign, or by pruning. We begin by examining the fine-tuning of linear layers through the lens of subspace analysis and define parameter interference as an optimization problem. Subsequently, we introduce an innovative approach called zero-shot Sparse Mixture of Low-rank Experts (SMILE) construction, which upscales source models into an MoE model without extra data or training. Our approach relies on the observation that fine-tuning mostly keeps the important parts from the pre-training, but it uses less significant or unused areas to adapt to new tasks. Additionally, the issue of parameter interference, which is intrinsically challenging in the original parameter space, can be managed by expanding the dimensions. We conduct extensive experiments across both image classification and text generation tasks, using full fine-tuning and LoRA fine-tuning, and we apply our method to LLMs, highlighting the adaptability and scalability of SMILE. For full fine-tuned models, about 50% additional parameters can achieve around 98%-99% of the performance of eight individual fine-tuned ViT models, while for LoRA fine-tuned Flan-T5 models, maintaining 99%

performance with only 2% extra parameters. Code is available at https://github.com/tanganke/fusion_bench.

Index Terms—Mixture of Experts, Model Fusion, Subspace Decomposition, Large Language Model.

I. INTRODUCTION

IN RECENT years, the field of deep learning has witnessed an exponential growth in model sizes and dataset scales, making the training of large-scale deep models on extensive datasets increasingly cost-prohibitive, both in terms of financial resources and environmental impact [1], [2]. Deep model fusion techniques have emerged as a promising solution, allowing the integration of knowledge from pre-existing models without the need for extensive retraining [3], [4], [5]. By leveraging deep model fusion, researchers can significantly reduce computational costs while creating more robust and versatile models that combine the strengths of multiple sources.

Following the categorization in [6], we classify model fusion methods into three main categories: model ensemble methods, model merging methods, and model mixing methods. Model ensemble techniques aggregate the predictions from several models to enhance performance [7], [8]. While resource-intensive in terms of memory and computation, it improves knowledge distillation training [9], [10]. Model merging methods, on the other hand, combine the parameters of multiple models into a single model, often through weighted averaging or parameter alignment [11], [12]. Model mixing methods involve the integration of multiple models through gating mechanisms or depth concatenation, allowing for more flexible and adaptive fusion strategies [13], [14]. These methods are particularly effective in multi-task learning scenarios, where the merged model can simultaneously perform multiple tasks.

However, despite the promising advancements in model fusion, several critical challenges persist, hindering the full realization of its potential. A primary concern is parameter interference between models, which can lead to suboptimal performance. Additionally, the lack of interpretability in the fusion process remains a significant hurdle, as current insights are largely confined to heuristic observations or simplified assumptions, such as linear mode connectivity, parameter signs or importance [15], [16], [17], [18]. Understanding how parameters are merged is crucial for building trust in the merged models and for further improving fusion techniques. These challenges are particularly pronounced in complex, high-dimensional, non-linear model

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Anke Tang, Yong Luo, Lefei Zhang, and Bo Du are with the School of Computer Science, National Engineering Research Center for Multimedia Software and Hubei Key Laboratory of Multimedia and Network Communication Engineering, Wuhan University, Wuhan 430072, China (e-mail: anketang@whu.edu.cn; yluo180@gmail.com; zhanglefei@whu.edu.cn; dubo@whu.edu.cn).

Li Shen is with the Shenzhen Campus of Sun Yat-sen University, Shenzhen 518107, China (e-mail: mathshenli@gmail.com).

Shuai Xie is with JD Explore Academy, Beijing 100176, China (e-mail: xieshuai@jd.com).

Han Hu is with the School of Information and Electronics, Beijing Institute of Technology, Beijing 100811, China (e-mail: hhu@bit.edu.cn).

Dacheng Tao is with Nanyang Technological University, Singapore 639798 (e-mail: dacheng.tao@ntu.edu.sg).

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architectures, where the interactions between parameters can be extremely intricate and non-intuitive.

Instead of relying on heuristic methods or simplified assumptions, we introduce a novel subspace perspective in this study to better understand and address the parameter interference problem. We first examine the fine-tuning process in linear layers through the lens of subspace analysis using matrix decomposition in Section II. This allows us to decompose the prediction of a fine-tuned model into distinct components, encompassing the pre-trained knowledge and task-specific adaptation. This approach provides insights into how models adapt to downstream tasks while preserving pre-trained knowledge. Building on these experimental observations, we develop a more comprehensive understanding of the fine-tuning process. In Section III, we further conceptualize parameter interference as an optimization problem, offering a more rigorous and quantifiable approach to addressing this challenge.

Based on our insights, we introduce an innovative approach called zero-shot Sparse Mixture of Low-rank Experts (SMILE) construction, enhancing existing source models into a more versatile sparse MoE model. A particularly significant feature of our approach is its zero-shot adaptability, as it facilitates the immediate deployment of fused models in new environments or tasks, drastically minimizing the time and resources typically required for model adaptation.

The effectiveness of our proposed method is rooted in two key observations derived from our subspace analysis. Firstly, we found that the fine-tuning largely preserves the most important pre-trained weights and primarily utilizes less significant or previously unused dimensions of the parameter space to adapt to new tasks. This preservation ensures that the critical pre-training knowledge encoded in the original models is not lost during fine-tuning and implies that the parameter subspace required to accommodate new knowledge may vary from task to task. Secondly, we found that while parameter interference is inherently difficult to address in the original parameter space, it becomes more manageable when we increase the model's dimensionality. This expansion creates additional 'room' for task-specific parameter updates to coexist without mutual interference.

We conducted extensive experiments across various tasks and models in both the vision and language domains, utilizing traditional full fine-tuning as well as Low-Rank Adaptation (LoRA) fine-tuning [19]. The experimental results show that for models that undergo full fine-tuning, adding approximately 50% more parameters allows us to achieve around 98-99% of the performance of eight individual fine-tuned models. In the case of LoRA fine-tuned models, maintaining 99% of the individual performance requires only a 2% increase in parameters. This method also offers trade-offs between performance and model size, as illustrated in Fig. 1, where we vary the rank k of local experts.

To summarize, our contributions in this study are as follows:

- We provide a novel subspace perspective on the fine-tuning process, shedding light on how models adapt to new tasks while preserving pre-trained knowledge. In addition, we formulate the parameter interference problem as an optimization problem, providing a more rigorous and measurable perspective on this issue.

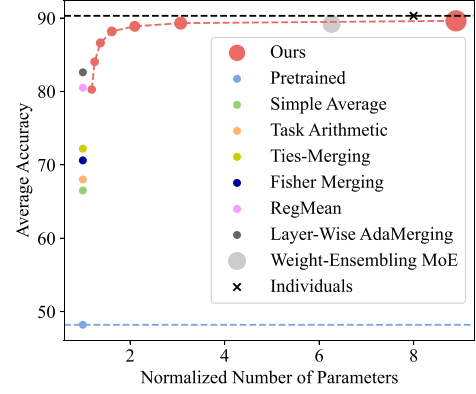


Fig. 1. Multi-task model fusion experiment on eight image classification tasks using CLIP-ViT-B/32 models. Here we set $k_{gate} = 16$ and k is varied from 4 to 128 to investigate the trade-off between performance and model size.

- We introduce a zero-shot Sparse Mixture of Low-Rank Experts (SMILE) construction approach, enabling the fusion of existing models into more unified versatile SMILE models. We also discuss the complexity of our method, highlighting its potential for broader applications in deep learning research and practice.
- We demonstrate the effectiveness of our method through extensive experiments on a variety of tasks and setups, showcasing its superior performance and efficiency compared to existing model fusion techniques.

II. RETHINKING MODEL FINE-TUNING FROM A SUBSPACE PERSPECTIVE

In this study, our goal is to create a unified, versatile model that integrates the strengths of multiple fine-tuned models, enabling it to perform various tasks or process inputs from different domains simultaneously. We denote the total number of fine-tuned models as T . Before we delve into the proposed method's details, we gain insights into the fine-tuning process from a singular value decomposition (SVD) subspace perspective. In this section, we aim to (1) investigate and locate the task information in the fine-tuned weights W_{ft} , and (2) understand how it is related to the pre-trained weights W .

Consider a linear layer of the pre-trained model with weight matrix $W \in \mathbb{R}^{m \times n}$ and bias vector $b \in \mathbb{R}^m$. After full fine-tuning on a downstream task, the weight matrix and bias vector are updated to W_{ft} and b_{ft} , respectively. To achieve a deeper understanding of these updates, we need to employ mathematical tools that allow us to decompose the parameter space into distinct ranges of importance, i.e. subspaces. We state the following theorem.

Theorem 1: Given two sets of orthonormal vectors $\{u_i\}_{i=1}^p \subset \mathbb{R}^m$ and $\{v_j\}_{j=1}^q \subset \mathbb{R}^n$, $1 \leq p \leq m$ and $1 \leq q \leq n$, the set of matrices $\{u_i v_j^T\}_{i=1, j=1}^{p, q}$ forms an orthonormal basis for a subspace of $\mathbb{R}^{m \times n}$ with dimension pq .

Proof: For simplicity, let $x_{ij} = u_i v_j^T$. The Frobenius inner product of two matrices x_{ab} and x_{cd} is defined as follows:

$$\langle x_{ab}, x_{cd} \rangle = \text{tr}(u_a v_b^T (u_c v_d^T)^T) = \text{tr}(u_a v_b^T v_d u_c^T) \in \mathbb{R}. \quad (1)$$

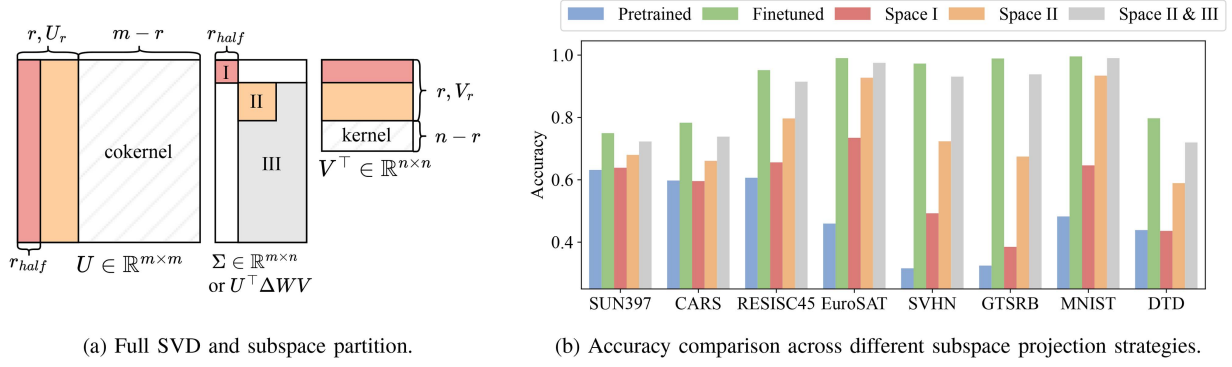


Fig. 2. This figure illustrates SVD decomposition and subspace partition of the singular value matrix Σ , along with an accuracy comparison of different subspace projection strategies discussed in Section II.

Orthonormality: we consider three cases:

- 1) If $a = c$ and $b = d$, then $\langle x_{ab}, x_{cd} \rangle = \text{tr}(u_a u_a^T) = u_a^T u_a = 1$.
- 2) If $b \neq d$, then $v_b^T v_d = 0$ and $\langle x_{ab}, x_{cd} \rangle = 0$.
- 3) If $b = d$ and $a \neq c$, then $v_b^T v_d = 1$ and $\langle x_{ab}, x_{cd} \rangle = \text{tr}(u_a u_c^T) = u_a^T u_c = 0$.

Thus, $\{x_{ij}\}_{i,j=1}^{p,q}$ is orthonormal.

Linear Independence: assume there exists a nonzero matrix $\alpha \in \mathbb{R}^{p \times q}$ such that $\sum_{i=1}^p \sum_{j=1}^q \alpha_{ij} x_{ij} = 0$. For any $a \in [p]$ and $b \in [q]$, take the inner product of both sides with x_{ab} . We obtain the following:

$$\left\langle \sum_{i=1}^p \sum_{j=1}^q \alpha_{ij} x_{ij}, x_{ab} \right\rangle = \langle 0, x_{ab} \rangle = 0. \quad (2)$$

By the linearity of the inner product and the orthogonality:

$$\sum_{i=1}^p \sum_{j=1}^q \alpha_{ij} \langle x_{ij}, x_{ab} \rangle = \alpha_{ab} \langle x_{ab}, x_{ab} \rangle = \alpha_{ab} = 0. \quad (3)$$

Since this holds for any a and b , we conclude that all $\alpha_{ij} = 0$. This leads to a contradiction to the assumption that α is nonzero. Therefore, $\{x_{ij}\}_{i,j=1}^{p,q}$ is linearly independent, which is the necessary and sufficient conditions for a set of elements to form a basis for a vector space with dimension pq . $\square$

We start by decomposing the weight matrix W using reduced SVD as $W = U_r \Sigma_r V_r^T$, where $U_r \in \mathbb{R}^{m \times r}$ and $V_r \in \mathbb{R}^{n \times r}$ are orthonormal matrices containing the left singular vectors and right singular vectors, respectively, $\Sigma_r \in \mathbb{R}^{r \times r}$ is a diagonal matrix containing the singular values sorted in descending order, and r is the rank of the matrix W [20]. In the case of full SVD, the matrices are $U \in \mathbb{R}^{m \times m}$, $\Sigma \in \mathbb{R}^{m \times n}$, and $V \in \mathbb{R}^{n \times n}$, which preserve all information about the matrix W , including its kernel (null space) and cokernel (left null space), as shown in Fig. 2(a).

Remark 1: According to Theorem 1, the set of matrices $\{u_i v_j^T | i \in [m], j \in [n]\}$ forms an orthonormal basis for a subspace of $\mathbb{R}^{m \times n}$ with dimension mn . In other words, for any real matrix $A \in \mathbb{R}^{m \times n}$, we can express it as a weighted sum of the elements in the basis, i.e. $A = \sum_{i=1}^m \sum_{j=1}^n \langle A, u_i v_j^T \rangle u_i v_j^T \in \text{span}(\{u_i v_j^T\}_{i,j=1}^{m,n})$.

Remark 2: Let $\mathcal{U}$ and $\mathcal{V}$ be two subsets of $\{u_i\}_{i=1}^m$ and $\{v_i\}_{i=1}^n$, respectively. $\{u v^T | u \in \mathcal{U}, v \in \mathcal{V}\}$ forms an orthonormal basis for a subspace of $\mathbb{R}^{m \times n}$, with dimension $|\mathcal{U}| |\mathcal{V}| \leq mn$.

To gain insights into how fine-tuning modifies the pre-trained weights to adapt them to a specific task, we assume the fine-tuned linear layer accepts an input $x \in \mathbb{R}^n$ and outputs $y = W_{ft} x + b_{ft}$. Because the row space $\{v_i\}_{i=1}^n$ is an orthonormal basis for $\mathbb{R}^n$, we can decompose x as $x = \sum_{i=1}^n \langle x, v_i \rangle v_i$, where $\langle \cdot, \cdot \rangle$ denotes the vector inner product. On the other hand, W_{ft} and b_{ft} are updated from the pre-trained weights W and b . We can derive the following equation:

$$y = W_{ft} x + b_{ft} \quad (4)$$

$$= (W + \Delta W) x + b + \Delta b \quad (5)$$

$$= \underbrace{W x + b}_{\text{pre-trained part}} + \underbrace{\Delta W x + \Delta b}_{\text{fine-tuned part}}. \quad (6)$$

Now we expand the pre-trained part and fine-tuned part in Eq.(4) separately as follows:

$$\text{pre-trained part} = \sum_{i=1}^n W \langle x, v_i \rangle v_i + b \quad (7)$$

$$= \sum_{i=1}^n \left(\sum_{j=1}^r \sigma_j u_j v_j^T \right) \langle x, v_i \rangle v_i + b \quad (8)$$

$$= \sum_{j=1}^r \sigma_j \langle x, v_j \rangle u_j + b, \quad (9)$$

$$\text{fine-tuned part} = \sum_{i=1}^n (W_{ft} - W) \langle x, v_i \rangle v_i + \Delta b \quad (10)$$

$$= \sum_{i=1}^n \left(\sum_{j=1}^m \sum_{k=1}^n \delta_{jk} u_j v_k^T \right) \langle x, v_i \rangle v_i + \Delta b \quad (11)$$

$$= \sum_{j=1}^m \sum_{k=1}^n \delta_{jk} \langle x, v_k \rangle u_j + \Delta b. \quad (12)$$

Where $\delta_{jk} = \langle \Delta W, u_j v_k^\top \rangle = (U^\top \Delta W V)_{jk}$ is the Frobenius inner product between the fine-tuned weight update ΔW and the rank-one matrix $u_j v_k^\top$. It also quantifies how much the weight updates align with the direction specified by $u_j v_k^\top$ and indicates which input-output transformation is enhanced or suppressed (or enhanced reversely) during fine-tuning, based on its sign and magnitude. For example, a large positive δ_{jk} indicates that the connection between the k -th input direction (v_k) and j -th output direction (u_j) is strengthened for the downstream task. This decomposition shows how the contributions of the pre-trained and fine-tuned components differ in shaping the output. Up to this point, we only understand that the fine-tuned update ΔW potentially uses all mn dimensions, while the pre-trained part only uses r dimensions.

We further split left/right singular vectors into three distinct subsets based on the distribution of the singular values of ΔW , and design an ablation study corresponding to different subspaces in the projection coefficient matrix $U^\top \Delta W V$: (1) The top-left subspace I contains the most significant singular values that cumulatively account for 50% of the total sum of the singular values, we denote the number of singular values in this subspace as r_{half} , $\sum_{i=1}^{r_{half}} \sigma_i \approx \sum_{i=1}^r \sigma_i / 2$. This subspace is crucial for preserving pre-training information. (2) The middle subspace II encompasses the singular values that make up the remaining half of the cumulative sum. These values are still important but less so than those in subspace I. (3) subspace III contains no information about the pre-trained weights, as its range is beyond $\text{rank}(W)$. This subspace partition is illustrated as the Σ in Fig. 2. We fine-tune the pre-trained CLIP-ViT-B/32 model on eight downstream tasks from different domains, including hand-written digit images, satellite images, regular patterns, car images, and natural images. Then we project the ΔW onto the different subspaces, including (1) subspace I $\text{span}(\{u_i v_j^\top | i \in [r_{half}], j \in [r_{half}]\})$, (2) subspace II $\text{span}(\{u_i v_j^\top | i \in [r_{half}, r], j \in [r_{half}, r]\})$, and (3) subspace II & III $\text{span}(\{u_i v_j^\top | i \in [r_{half}, m], j \in [r_{half}, n]\})$. We compare the performance of the pre-trained model, the fine-tuned models, and modified fine-tuned models with different subspace projection strategies applied on ΔW in Fig. 2(b). For more details and additional quantitative analysis about the projected energy of ΔW , please refer to Appendix A.

Fig. 2(b) demonstrates that projecting fine-tuned updates onto subspace I result in a slight improvement in performance on downstream tasks compared to the pre-trained model, sometimes showing no improvement at all. Projection onto subspace II leads to moderate improvement, while projection onto subspace II & III results in significant performance gains, nearly reaching the level of the fine-tuned model. Based on these observations, we draw the following conclusions:

Fine-tuning maintains the most important pre-trained features, but leverages less significant dimensions for task-specific learning and activates or repurposes previously unused dimensions in the weight space.

This suggests that the significant components of the fine-tuned updates ΔW are concentrated in directions orthogonal or complementary to the dominant pre-trained features. This insight motivates the use of low-rank approximations of ΔW , which can efficiently capture these task-specific updates without interfering with the shared pre-trained knowledge. Recent work has further highlighted the minimal effective changes required during fine-tuning, suggesting that fine-tune updates may be highly compressible or low-rank in nature [19], [21].

III. PARAMETER INTERFERENCE BETWEEN TASK-SPECIFIC MODELS

From the previous section, we build an understanding of how fine-tuning modifies the pre-trained weights to adapt to a specific downstream task. In this section, we investigate the parameter interference between models fine-tuned on different tasks, which has been widely explored in multi-task learning and multi-task model merging, primarily within the model parameter space. We add superscripts to denote the task index, e.g. $W_{ft}^{(i)}$ and $b_{ft}^{(i)}$ for the i -th task.

Assume we have T tasks, each associated with its own fine-tuned model. In the simplest cases, we focus on the linear layers, where a common input x is processed to produce T different outputs: $y^{(1)}, y^{(2)}, \dots, y^{(T)}$. According to Eq.(4) and Eq.(12), each $y^{(i)}$ can be decomposed into the pre-trained part and fine-tuned part as follows:

$$y^{(i)} = \underbrace{\sum_{j=1}^r \sigma_j \langle x, v_j \rangle u_j + b}_{\text{pre-trained part}} + \underbrace{\sum_{j=1}^m \sum_{k=1}^n \delta_{jk}^{(i)} \langle x, v_k \rangle u_j + \Delta b^{(i)}}_{\text{fine-tuned part}}. \quad (13)$$

In this formulation, the pre-trained term is shared and remains constant during the fine-tuning process across all tasks. When it comes to model merging, these fine-tuned models are merged to construct a unified multi-task model capable of handling all tasks simultaneously. A common method for merging involves taking a weighted average of the fine-tuned weights, i.e. $W_{merged} = \sum_{l=1}^T \lambda_l W_{ft}^{(l)}$ and $b_{merged} = \sum_{l=1}^T \lambda_l b_{ft}^{(l)}$. This is equivalent to merging the fine-tuned parts of the models, while the pre-trained parts are shared across all tasks. Therefore, we express the output of the merged model as:

$$y_{merged} = \text{pre-trained part} + \sum_{j=1}^m \sum_{k=1}^n \left(\sum_{l=1}^T \lambda_l \delta_{jk}^{(l)} \right) \langle x, v_k \rangle u_j + \sum_{l=1}^T \lambda_l \Delta b^{(l)}. \quad (14)$$

Substitute the input x with $x^{(i)}$ from the i -th task (domain), we aim to minimize the discrepancy between the output of the merged model and the output of the i -th fine-tuned model. We formulate the optimization problem as follows:

$$\min_{\lambda_l} \left\| y_{merged} - y^{(i)} \right\|_2^2 \quad (15)$$

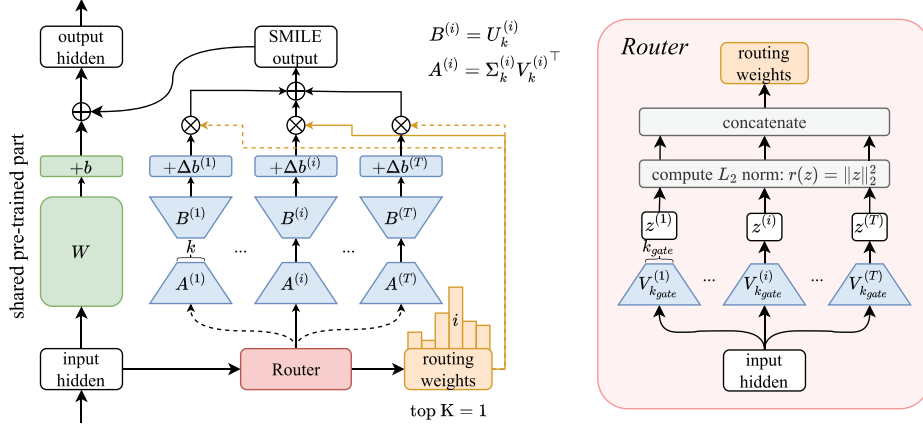


Fig. 3. The architecture of the proposed Sparse Mixture of Low-rank Experts (SMILE) model.

$$= \min_{\lambda_l} \left\| \sum_{j=1}^m \sum_{k=1}^n \left[\left(\sum_{l=1}^T \lambda_l \delta_{jk}^{(l)} \right) - \delta_{jk}^{(i)} \right] \langle x^{(i)}, v_k \rangle u_j \right\|_2^2 \quad (16)$$

$$+ \left\| \left(\sum_{l=1}^T \lambda_l \Delta b^{(l)} \right) - \Delta b^{(i)} \right\|_2^2. \quad (17)$$

Eq.(15-17) formulates the optimization problem for weight interpolation-based model merging such as Task Arithmetic. Using the triangle inequality, we can decompose the error item into two parts and assert an upper bound as follows:

$$\begin{aligned} & \left\| y_{merged} - y^{(i)} \right\|_2^2 \\ & \leq \underbrace{\left\| \sum_{j=1}^m \sum_{k=1}^n \left[\left(\sum_{l=1}^T \lambda_l \delta_{jk}^{(l)} \right) - \delta_{jk}^{(i)} \right] \langle x^{(i)}, v_k \rangle u_j \right\|_2^2}_{\text{weight term}} \\ & \quad + \underbrace{\left\| \left(\sum_{l=1}^T \lambda_l \Delta b^{(l)} \right) - \Delta b^{(i)} \right\|_2^2}_{\text{bias term}}. \end{aligned} \quad (18)$$

For the bias term, we have a closed-form solution for $\lambda_l = (\Delta B^\top \Delta B)^{-1} \Delta B \Delta b^{(i)}$, where ΔB is a matrix with l -th columns as $\Delta b^{(l)}$. Notice that this solution varies with the task (domain) index of the coming input $x^{(i)}$, so a more straightforward solution is to fix the bias term during the fine-tuning process, so that $\Delta b^{(i)} = 0$ for all i . As for the weight term, parameter interference does not occur on subspace that is orthogonal to the input, i.e. $\text{span}(\{u_i v_j^\top | i \in [m], j \in [n] \text{ and } \langle x^{(i)}, v_j \rangle = 0\})$, thus a possible strategy is to enlarge the input size to increase the dimension of the orthogonal subspace. This explains why model merging methods perform better on larger models with more dimension redundancy. When the input activates certain dimensions, i.e. for k such that $\langle x^{(i)}, v_k \rangle \neq 0$, the interference is inevitable unless the domain gap between different tasks is large enough to make the activation dimensions disjoint. Note that we can gain the same conclusion within the original model

parameter space, by simply replacing the basis vectors $\{u_i\}_i$ and $\{v_i\}_i$ in this section with the standard Euclidean basis vectors $\{e_i\}_i$.

IV. RESOLVING PARAMETER INTERFERENCE USING SPARSE MIXTURE OF LOW-RANK EXPERTS

Recognizing that addressing parameter interference by model merging is challenging, particularly for bias terms, we note that the optimal weight combination method has a closed-form solution that varies by task. To manage this challenge, we introduce an innovative approach with a Sparse Mixture of Low-rank Experts (SMILE) model in this section, which operates in a zero-shot fashion, meaning no data or training is required. An overview is shown in Fig. 3.

We upscale the linear layers from source models to the SMILE model, which consists of a shared pre-trained part, a router, and several low-rank experts. Fig. 3 is organized into two primary sections: the overall model architecture (left) and the routing mechanism (right).

Recall the output decomposition in Eq.(4) and Eq.(14), we can express the output of a merged model as the output of the pre-trained model plus a weighted sum of the fine-tuned parts of the individual models. If we can identify the most relevant experts for a given input, we can dynamically select the corresponding fine-tuned parts to combine with the pre-trained part. Then the merging error in Eq.(15) can be minimized. Mathematically, we can express this idea as:

$$\begin{aligned} y_{merged} &= \text{pre-trained part} \\ &+ \sum_{j=1}^m \sum_{k=1}^n \left(\sum_{l=1}^T \lambda_l \left(x^{(i)} \right) \delta_{jk}^{(l)} \right) \langle x^{(i)}, v_k \rangle u_j \\ &+ \sum_{l=1}^T \lambda_l \left(x^{(i)} \right) \Delta b^{(l)}. \end{aligned} \quad (19)$$

Here, λ is a function that maps the input to a one-hot probability distribution over the tasks, i.e. $\lambda_j(x^{(i)}) = 1$ if $j = i$, and $\lambda_j(x^{(i)}) = 0$ otherwise. However, a naive implementation of this idea would require a training process to learn the parameters of

the router and a large number of additional parameters to store the fine-tuned weights of all tasks. A more efficient approach is to remove less significant terms from the fine-tuned components in Eq.(19), focusing on retaining the most pertinent knowledge for each task. Therefore, the parameter space must be ranked by the importance of its dimensions. However, from previous findings in Section II, we know that the fine-tuned information is distributed across less significant dimensions (Space II & III), which is a large portion of the whole space. We opt to use SVD to decompose the parameter differences $\Delta W^{(i)}$ for each task, and then apply a low-rank approximation to extract the most important part as follows:

$$\Delta W^{(i)} = U^{(i)} \Sigma^{(i)} V^{(i)\top} \quad (20)$$

$$= \sum_{j=1}^{r^{(i)}} \sigma_j^{(i)} u_j^{(i)} v_j^{(i)\top} \approx \sum_{j=1}^k \sigma_j^{(i)} u_j^{(i)} v_j^{(i)\top} \quad (21)$$

$$= U_k^{(i)} \Sigma_k^{(i)} V_k^{(i)\top}, 1 \leq k \leq r^{(i)}. \quad (22)$$

Where $r^{(i)}$ is the rank of the fine-tuned weight matrix $\Delta W^{(i)}$, and k is the rank of the low-rank approximation, which is determined as a hyperparameter. $U_k^{(i)}$ and $V_k^{(i)}$ contains the first k columns of $U^{(i)}$ and $V^{(i)}$, respectively. Here we drop the terms with indices $j > k$ in the summation, which correspond to the less significant dimensions. Let $A^{(i)} = \Sigma_k^{(i)} V_k^{(i)\top}$ and $B^{(i)} = U_k^{(i)}$, we can express the approximation similar to a LoRA adapter: $\Delta W x = B^{(i)} A^{(i)} x$. The following theorem states the optimality of this low-rank approximation.

Theorem 2: Given a matrix $W \in \mathbb{R}^{m \times n}$, its low-rank approximation $W_k = U_k \Sigma_k V_k^\top$ with rank k minimizes the Frobenius norm of the difference between W and W_k , i.e. $W_k = \arg \min_{\text{rank}(W')=k} \|W - W'\|_F$.

Another critical component of the SMILE model is the router, which determines the routing weights. The routing weights should reflect the importance of each expert for a given input, and we hypothesize that the most important dimensions of the parameter updates are more likely to align with the input vector. This hypothesis is supported by an analysis of the gradient flow during fine-tuning [22], [23], which we discuss in detail in Appendix B. Consequently, we design the routing logits as the L_2 norm of the projections of the input onto the low-rank matrices. The routing weights can be expressed as follows:

$$r^{(i)} = \left\| \sum_{j=1}^{k_{gate}} \langle x, v_j^{(i)} \rangle \right\|_2 = \|V_{k_{gate}}^{(i)\top} x\|_2. \quad (23)$$

Where k_{gate} is the number of dimensions used for routing, which is a hyperparameter. k_{gate} should not be excessively large, which could diminish the distinctiveness of routing weights. In the extreme case where $k_{gate} = n$, $r^{(i)} = \|x\|_2$, which is equivalent to a uniform distribution over all experts. In our hyperparameter analysis, we find that $k_{gate} = 4$ or 8 is a good choice for most tasks. To summarize, the output of the SMILE module can be

expressed as follows:

$$y = (Wx + b) + \sum_{i=1}^T \frac{\lambda_i}{\sum_{j=1}^T \lambda_j} \left(U_k^{(i)} \Sigma_k^{(i)} V_k^{(i)\top} x + \Delta b^{(i)} \right) \quad (24)$$

$$\lambda_i = \begin{cases} p_i, & p_i \in \text{TopK}(\{p_j\}_{j=1}^T, K) \\ 0, & \text{otherwise} \end{cases} \quad (25)$$

$$p_i = \text{softmax}_i(r^{(i)}) = \text{softmax}_i \left(\|V_{k_{gate}}^{(i)\top} x\|_2 \right). \quad (26)$$

The potential accumulation of approximation errors across layers in SMILE, due to its layer-wise low-rank approximations of $\Delta W^{(i)}$, is mitigated through several design choices: 1) By tuning the hyperparameter k , SMILE balances approximation accuracy and efficiency; 2) The dynamic router prioritizes relevant experts, minimizing the impact of less accurate updates, while shared pre-trained weights provide a robust foundation, limiting error propagation. Empirical results, such as 97.1-98.9% normalized average accuracy on eight CLIP-ViT-B/32 models, demonstrate that these mechanisms effectively control error accumulation, ensuring high performance with minimal overhead. In Appendix F-3, we ablate the impact of the low-rank approximation and the routing mechanism.

Complexity Analysis: The linear layer has $m(n+1)$ parameters. The upscaled SMILE module has $m(n+1) + T(mk + nk + m) + nTk_{gate}$ parameters, the additional parameters have a space complexity of $O(T(mk + n(k + k_{gate})))$. For every input token, an additional number of parameters of $nTk_{gate} + K(mk + nk + m)$ are activated, with K representing the top- K hyperparameter. For instance, with $T = 8$, $k_{gate} = 4$, $k = 32$, $K = 1$ and $m = n = 1024$, the SMILE model has $565K$ additional parameters, which is about 53.9% of the original parameter count. 99K additional parameters are activated for each input token, which is only about 9.4% of the original parameter count.

Extending to Parameter-efficient fine-tuned (PEFT) models: It is straightforward to extend the SMILE upscaling to PEFT models, such as LoRA fine-tuned models. We can still decompose the fine-tuning updates using SVD as $\Delta W_{LoRA} = B_{LoRA} A_{LoRA}$. Note that for parameter-efficient fine-tuned models, such as LoRA fine-tuned models, k_{gate} should be set to a smaller value than the hyperparameter rank of LoRA r_{LoRA} , and $k \leq r_{LoRA}$. This approach complements recent advancements in modular LLMs, where libraries of LoRA adapters are reused for training-free model merging [24].

V. EXPERIMENTS

In this section, we evaluate the effectiveness of the proposed SMILE on a variety of setups, including image classification [25], [26], [27], [28], [29], [30], [31], [32] and text generation tasks, as well as full fine-tuning and LoRA fine-tuning. Detailed information about the fine-tuned models is in Appendix C. We compare our method with several SOTA model fusion techniques, including Simple Averaging [33], Fisher merging [11], RegMean [12], Task Arithmetic [34], Ties-Merging [17], AdaMerging [35], Representation Surgery [36],

TABLE I
REQUIREMENTS OF DIFFERENT MODEL FUSION METHODS

Methods	Validation Set	Test-Time Adaptation
Weight Averaging	×	×
Fisher-Merging	✓	×
RegMean	✓	×
Task Arithmetic	✓	×
Ties-Merging	✓	×
AdaMerging	×	✓
AdaMerging + Surgery	×	✓
TSVM	✓	×
ISO-C	✓	×
WEMoE	×	✓
SMILE (Ours)	×	×

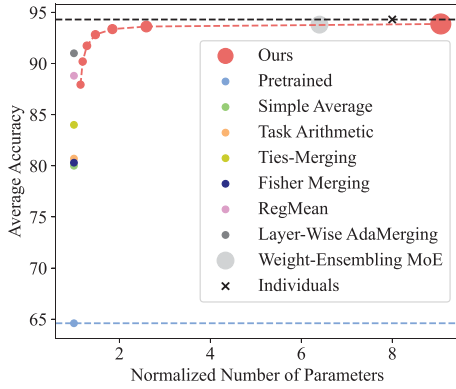


Fig. 4. Multi-task model fusion experiment on eight image classification tasks using CLIP-ViT-L/14 models ($k_{gate} = 16$).

TSVM [37], ISO-C [38] and WEMoE [39]. To further demonstrate the scalability of SMILE upscaling, we also conduct experiments using off-the-shelf large language models fine-tuned from Mistral-7B-v0.1. Our code implementation is based on FusionBench [6], a comprehensive benchmark and toolkit for model fusion.

A. Multi-Task Model Fusion on Open-Vocabulary Image Classification Tasks

We first evaluate our proposed SMILE method on eight diverse open-vocabulary image classification tasks using CLIP models from HuggingFace.^{1,2} For each task, the text encoder of the pre-trained model is frozen, and only the vision encoder is fine-tuned. Table I presents the requirements of different model fusion methods, highlighting that SMILE is a training-free model fusion method that does not require additional labeled samples or test-time adaptation.

Figs. 1 and 4 illustrates the average accuracy of the merged model across different methods, for SMILE k_{gate} is set to 16 and k is varied from 4 to 128. These two figures demonstrate the effectiveness of SMILE and the trade-off between performance and model size.

In Tables II and III, we compare the performance of various model fusion methods on CLIP-ViT-B/32 and CLIP-ViT-L/14 models, respectively. Our SMILE method achieves competitive results across all tasks, suggesting it effectively preserves task-specific adaptations despite domain diversity. For instance, with CLIP-ViT-L/14 models, SMILE ($k_{gate} = 16$, $k = 128$) achieves 99.3% of the individual model performance while using only 2.56 times the parameters of a single model, compared to eight individual fine-tuned models and Weight-Ensembling MoE, which requires 6.40 times the parameters.

Besides, methods that directly interpolate weights tend to suffer from parameter interference, resulting in degraded performance. This is evident in Simple Averaging, which shows a substantial drop in average accuracy compared to individual models. In contrast, MoE-based methods such as WEMoE and SMILE effectively mitigate interference by isolating task-specific updates as experts, allowing them to maintain performance levels close to specialized individual models.

B. Multi-Task Model Fusion on Text Generation Tasks

We further evaluate SMILE on text generation tasks using Flan-T5-base models,³ which are fine-tuned on eight tasks from the GLUE benchmark [40]. We use two different fine-tuning strategies: full fine-tuning and LoRA fine-tuning with $r_{LoRA} = 16$. We present the results in Tables IV and V, for full fine-tuned models and LoRA fine-tuned models, respectively. For fully fine-tuned models, SMILE consistently outperforms other fusion methods across all eight tasks. With just 1.52 times the parameters of a single model, SMILE ($k_{gate} = 8$, $k = 32$) achieves 99.0% of the individual model performance with 1.52 times the parameters of a single model. In the LoRA fine-tuned scenario, SMILE maintains strong performance with minimal parameter increase (1.02 times). It achieves 99.3% of the individual model performance, significantly surpassing other multi-task model fusion methods. This is because LoRA’s low-rank updates inherently reduce interference by focusing on task-specific subspaces, and SMILE amplifies this by sparsely combining these updates. This suggests LoRA+SMILE as a preferable choice for balanced multi-task learning with limited resources.

C. Sparse Mixture of Low-Rank Experts Analysis

To better understand SMILE, we further conduct ablation studies using CLIP and Flan-T5 models.

1) *Ablations on the Low-Rank Approximation Rank k and Routing Dimension k_{gate} (Hyperparameter Analysis)*: Here we present a detailed analysis of the hyperparameters k and k_{gate} for both CLIP-ViT and Flan-T5 models. We also test the extreme cases when $k = \infty$, which corresponds to full-rank experts, denoted as “Dense” in the figures. The number of parameters is normalized by the number of parameters in the original model (87.5 M for CLIP-ViT-B/32 and 303 M for CLIP-ViT-L/14) to facilitate comparison. We examine the impact of hyperparameter configurations on both model performance (average accuracy)

¹<https://huggingface.co/openai/clip-vit-base-patch32>

²<https://huggingface.co/openai/clip-vit-large-patch14>

³<https://huggingface.co/google/flan-t5-base>

TABLE II

MULTI-TASK MODEL FUSION EXPERIMENTS ON EIGHT IMAGE CLASSIFICATION TASKS USING CLIP-ViT-B/32 MODELS. THIS TABLE PRESENTS TWO DIFFERENT HYPERPARAMETER SETTINGS FOR OUR METHOD: (1) $k_{gate} = 16$, $k = 32$ AND THE NORMALIZED PARAMETER COUNT IS 1.61; (2) $k_{gate} = 16$, $k = 128$ AND THE NORMALIZED PARAMETER COUNT IS 3.07

Method	SUN397	Cars	RESISC45	EuroSAT	SVHN	GTSRB	MNIST	DTD	Avg.
Individual	75.0	78.3	95.2	99.0	97.3	98.9	99.6	79.7	90.3 (100%)
Simple Averaging	65.4	62.6	70.8	76.9	64.5	54.9	86.3	50.9	66.5 (73.6%)
Fisher Merging	66.7	64.0	72.2	91.6	69.0	64.3	83.5	53.7	70.6 (78.2%)
RegMean	67.8	68.9	82.5	94.4	90.6	79.2	97.6	63.2	80.5 (89.1%)
Task Arithmetic	57.1	55.7	64.9	76.7	77.9	68.5	96.1	47.2	68.0 (75.3%)
Ties-Merging	67.1	64.2	74.1	76.8	77.7	69.4	94.1	54.0	72.2 (80.0%)
AdaMerging	67.9	71.3	83.5	92.7	87.4	92.9	98.2	67.0	82.6 (91.5%)
AdaMerging + Surgery	70.4	73.4	89.7	95.8	90.8	96.3	98.7	74.5	86.2 (95.5%)
TSVM	67.6	71.6	84.7	93.4	91.9	92.5	98.9	63.8	83.1 (92.0%)
ISO-C	69.7	72.9	86.0	89.8	87.6	92.4	98.5	65.4	82.8 (91.7%)
WEMoE ($\times 6.27$)	73.7	76.8	93.4	98.2	96.8	98.2	99.6	76.6	89.2 (98.8%)
SMILE (1, $\times 1.61$)	73.6	74.4	89.5	98.1	95.4	97.3	99.5	76.3	87.7 (97.1%)
SMILE (2, $\times 3.07$)	73.6	77.8	92.0	98.3	96.9	98.1	99.6	78.1	89.3 (98.9%)

TABLE III

MULTI-TASK MODEL FUSION EXPERIMENTS ON EIGHT IMAGE CLASSIFICATION TASKS USING CLIP-ViT-L/14 MODELS. THIS TABLE PRESENTS TWO DIFFERENT HYPERPARAMETER SETTINGS FOR OUR METHOD: (1) $k_{gate} = 16$, $k = 32$ AND THE NORMALIZED PARAMETER COUNT IS 1.47; (2) $k_{gate} = 16$, $k = 128$ AND THE NORMALIZED PARAMETER COUNT IS 2.56

Method	SUN397	Cars	RESISC45	EuroSAT	SVHN	GTSRB	MNIST	DTD	Avg.
Individual	82.8	92.9	97.4	99.2	97.9	99.2	99.8	85.5	94.3 (100%)
Simple Averaging	72.5	81.5	82.2	90.0	81.6	74.0	96.6	61.8	80.0 (84.8%)
Fisher Merging	70.6	79.4	84.1	98.1	74.7	85.0	89.5	61.0	80.3 (85.2%)
RegMean	75.3	88.4	90.0	97.1	95.9	92.4	98.5	72.6	88.8 (94.2%)
Task Arithmetic	72.0	79.0	80.5	86.0	87.5	83.5	98.0	58.8	80.7 (85.6%)
Ties-Merging	74.7	83.3	86.4	91.3	89.7	85.2	97.8	63.9	84.0 (89.1%)
AdaMerging	78.1	90.7	90.8	96.5	94.8	97.5	98.6	81.3	91.0 (96.5%)
AdaMerging + Surgery	79.3	91.5	93.4	96.9	95.2	97.9	98.8	83.2	92.0 (97.6%)
TSVM	78.2	89.8	93.5	96.7	95.6	96.5	99.1	75.3	90.6 (96.1%)
ISO-C	80.7	91.5	95.3	97.2	95.1	97.8	99.1	80.3	92.1 (97.7%)
WEMoE ($\times 6.40$)	81.5	92.3	96.5	98.8	97.6	99.4	99.6	84.5	93.8 (99.5%)
SMILE (1, $\times 1.47$)	79.9	91.0	94.3	99.0	97.9	98.6	99.7	82.2	92.8 (98.4%)
SMILE (2, $\times 2.56$)	81.9	92.3	95.5	99.1	98.0	98.9	99.7	83.6	93.6 (99.3%)

TABLE IV

MULTI-TASK PERFORMANCE WHEN MERGING FLAN-T5-BASE (FULL FINE-TUNED) MODELS ON ALL EIGHT TASKS. THIS TABLE PRESENTS TWO DIFFERENT HYPERPARAMETER SETTINGS FOR OUR METHOD: (1) $k_{gate} = 4$, $k = 16$ AND THE NORMALIZED PARAMETER COUNT IS 1.26; (2) $k_{gate} = 8$, $k = 32$ AND THE NORMALIZED PARAMETER COUNT IS 1.52

Method	CoLA	MNLI	MRPC	QNLI	QQP	RTE	SST2	STSB	Avg.
Individual	75.0	83.4	87.5	91.5	85.4	85.9	93.6	88.7	86.4 (100%)
Weight Averaging	69.1	62.6	79.4	89.8	83.9	81.2	91.7	73.2	78.9 (91.3%)
Task Arithmetic	70.5	57.8	78.4	90.2	83.6	80.5	92.3	77.8	78.9 (91.3%)
Ties-Merging	70.3	65.0	78.9	90.2	83.5	81.6	91.7	78.3	79.9 (92.5%)
SMILE (1, $\times 1.26$)	72.0	84.2	84.3	91.3	84.7	84.1	93.3	87.0	85.1 (98.5%)
SMILE (2, $\times 1.52$)	73.2	84.2	85.0	91.3	84.9	84.8	93.5	87.3	85.5 (99.0%)

TABLE V

MULTI-TASK PERFORMANCE WHEN MERGING FLAN-T5-BASE (LORA FINE-TUNED) MODELS ON ALL EIGHT TASKS. WE CHOOSE $k_{gate} = 2$, $k = 4$ AND THE NORMALIZED PARAMETER COUNT IS 1.02

Method	CoLA	MNLI	MRPC	QNLI	QQP	RTE	SST2	STSB	Avg.
Individual	69.1	82.7	85.5	90.9	84.0	84.4	92.9	87.4	84.6 (100%)
Weight Averaging	69.7	59.7	78.9	90.1	83.8	80.5	91.2	72.0	78.2 (92.4%)
Task Arithmetic	68.8	55.2	78.7	89.8	83.7	79.1	91.5	72.4	77.4 (91.5%)
Ties-Merging	68.3	56.3	79.4	89.8	83.7	79.4	91.6	71.2	77.5 (91.6%)
SMILE ($\times 1.02$)	69.3	82.9	83.8	90.6	83.9	83.4	93.1	85.1	84.0 (99.3%)

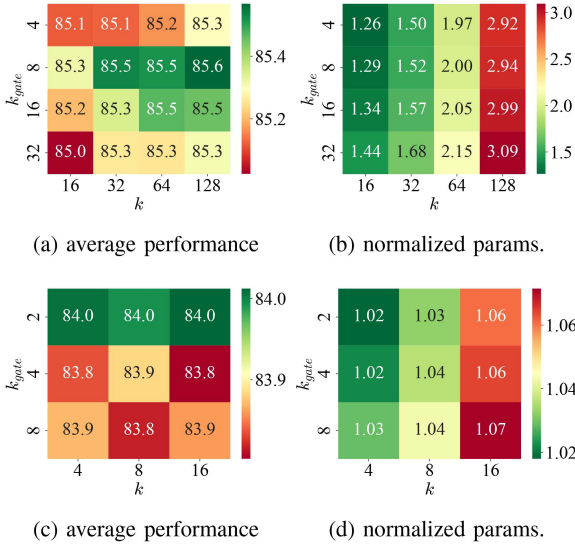


Fig. 5. Hyperparameter analysis of the Flan-T5-Base models on eight tasks from GLUE benchmark. We show how different values of hyperparameters k and k_{gate} affect the average performance and the normalized number of parameters in the upscaled model. Subfigures (a), and (b) show the results of the full fine-tuned models, while subfigures (c), and (d) show the results of the fine-tuned models with $r_{LoRA} = 16$.

and model complexity (number of parameters), and the experimental results demonstrate the flexibility and robustness of SMILE across different model architectures and tasks. Fig. 5 illustrates the impact of hyperparameters k and k_{gate} on performance and parameter count for Flan-T5-Base models in both full and LoRA fine-tuned scenarios. For CLIP-ViT models, Figs. 9 and 10 provide detailed heatmaps and line plots showing the relationship between hyperparameters, average accuracy, and parameter count. From Figs. 9 and 10, we observe that (1) The performance of the upscaled models is generally better than the pre-trained models, which demonstrates the effectiveness of our fine-tuning strategy. (2) increasing the values of k generally improves the performance of both the CLIP-ViT-B/32 and CLIP-ViT-L/14 models, though at the cost of increased model complexity. (3) Increased values of k_{gate} improve the performance of the upscaled models at the beginning, but the performance starts to decrease when k_{gate} is too large. This observation is consistent with our discussion in Section IV that a larger k_{gate} may result in a less discriminative gating mechanism. (4) Better performance preservation can be achieved with the CLIP-ViT-L/14 model than with the CLIP-ViT-B/32 model, which is consistent with our discussion in Section III that the larger model has more dimension redundancy and is less severe to the parameter interference problem.

In practice, when selecting hyperparameters for the upscaled models, it is crucial to balance the trade-off between performance and parameter overhead. Take CLIP-ViT-B/32 as an example, a good trade-off between performance and parameter overhead can be achieved with $k \approx 32$ and $k_{gate} \approx 16$. For the CLIP-ViT-L/14 model, $k \approx 64$ and $k_{gate} \approx 8$ are recommended. By doing so, we obtain a multi-task model that achieves around 98% of the performance of the fine-tuned model with only 20% of the total parameters compared to maintaining

eight individual fine-tuned models for each task. Note that the upscaled SMILE model is sparsely inferred, increasing the number of parameters by N only increases the activated parameters by about N/T for each token. Even with an extreme focus on storage and inference costs, only 7% of the parameters overhead can achieve an average multi-task performance of about 90% of the individual fine-tuned models.

Flan-T5 Models: Fig. 5 shows the hyperparameter analysis of the Flan-T5-Base models on eight tasks from the GLUE benchmark. We conduct the same analysis for both full fine-tuned models and LoRA fine-tuned models with $r_{LoRA} = 16$. It is observed that the performance is relatively stable, with most configurations yielding accuracies around 85.1 to 85.6 for the full fine-tuned models and around 83.8 to 84.0 for the LoRA fine-tuned models. Upscaling LoRA fine-tuned models is very parameter-efficient, with the number of parameters increasing by only 2% to 7% compared to the original dense model. For a balanced trade-off between performance and parameter overhead, consider setting $k \approx 32$ and $k_{gate} \approx 8$ for the full fine-tuned model fusion, and $k \approx 8$ and $k_{gate} \approx 2$ for the LoRA fine-tuned model fusion.

Across all upscaled SMILE models, we observe a consistent trend: increasing k and k_{gate} generally leads to improved performance, though the gains diminish as the parameter count increases. Notably, SMILE achieves near-optimal performance with relatively small values of k and k_{gate} . This analysis highlights the effectiveness of SMILE in balancing performance and efficiency, allowing users to fine-tune the trade-off based on their specific requirements. Furthermore, the stability of performance across a range of hyperparameter values also underscores the robustness of our method, making it well-suited for various multi-task fusion scenarios.

2) Ablations on Top-K Routing (Routing Analysis): Here we compare different values K in the top- K routing mechanism. The plots in Fig. 6 illustrate the impact of varying K on both the average accuracy across all tasks (Fig. 6a) and the accuracy of each individual task (Fig. 6b) when using the CLIP-ViT-B/32 model across eight image classification tasks. We observe that the average accuracy decreases slightly as K increases. This suggests that larger values of K , which allow more experts to be used for each input, are not necessary for multi-task model fusion where each expert is specialized for a specific task. In general, the performance of individual tasks is relatively stable across different values of K , indicating the robustness of the routing mechanism of SMILE (23).

D. Scalability to Large-Scale Models (Mistral-7B Models)

To demonstrate the scalability of SMILE to large-scale models, we conduct experiments on Mistral-7B models. We use Mistral-7B-v0.1 as our base pre-trained model, referred to as M_0 , and acquire three specialized models from HuggingFace [41]. The expert models are respectively labeled as M_1 , M_2 , and M_3 . To summarize, we have four models:

- Base pre-trained model (M_0): *mistralai/Mistral-7B-v0.1*
- M_1 : *meta-math/MetaMath-Mistral-7B*
- M_2 : *cognitivecomputations/dolphin-2.1-mistral-7b*
- M_3 : *uukuguy/speechless-code-mistral-7b-v1.0*

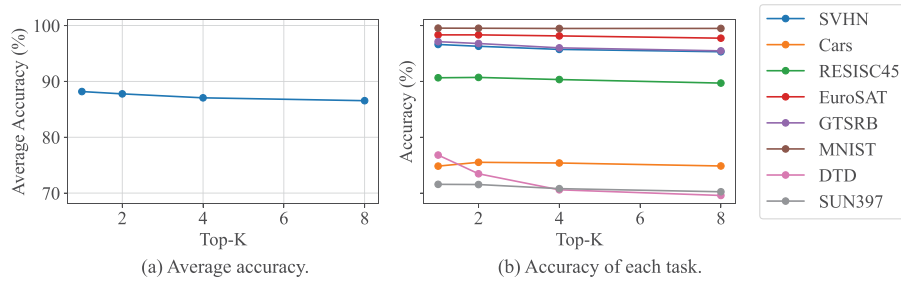


Fig. 6. Ablations on the Top- K routing for CLIP-ViT-B/32 models on eight image classification tasks ($k_{gate} = 16, k = 32$). Here we show the average accuracy and the accuracy on each task, and the y-axis is shared.

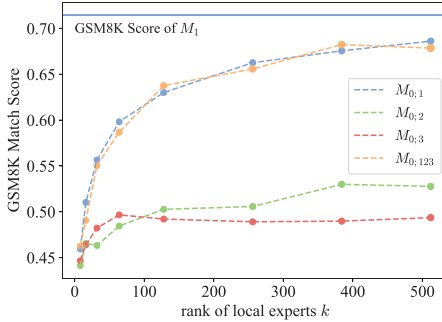


Fig. 7. The GSM8K benchmark scores of upscaled Mistral-7B models with varying k .

TABLE VI

COMPARISON OF INDIVIDUAL MISTRAL-7B MODELS AND THE UPSCALED MODEL ON VARIOUS BENCHMARK TASKS. FOR OUR METHOD, WE SET $k_{gate} = 8, k = 512$, AND THE TOTAL PARAMETER COUNT IS 11.2B. FOR A BETTER COMPARISON, WE ALSO INCLUDE THE PERFORMANCE OF THE Qwen1.5-14B MODEL AS A REFERENCE

Model	MMLU	TruthfulQA	GSM8K	ARC Challenge
M_0 (pre-trained)	59.64	42.62	38.81	53.92
M_1	60.56	44.79	71.49	51.02
M_2	60.56	55.88	56.93	57.00
M_3	61.18	47.47	48.98	57.68
MergeKit-MoE [42]	60.05	46.36	63.84	54.52
$M_{0;123}$ (Ours)	60.66	52.79	67.85	54.35
Qwen1.5-14B	66.11	52.00	69.37	49.93

Of these models, M_1 stands out as the sole expert in mathematics. To demonstrate the efficacy of the zero-shot routing mechanisms, we construct four distinct series of SMILE models with various expert combinations. These series are designated as $M_{0;1}$, $M_{0;2}$, $M_{0;3}$, and $M_{0;123}$, with $M_{0;i_1 \dots i_n}$ indicating the SMILE model that combines M_0 with the expert models $M_{i_1}, \dots, M_{i_n}$.

In Fig. 7, we present the GSM8K benchmark scores [43] of the upscaled Mistral-7B models with varying rank of local experts k with a constant rank of routers $k_{gate} = 8$. This plot highlights the trade-offs in selecting expert rank k for the upscaled SMILE model, where the GSM8K score generally improves as k increases, but this improvement is more pronounced for specific expert combinations, particularly for $M_{0;1}$ and $M_{0;123}$. This suggests that the routers succeed in selecting the proper expert for math problems. Table VI presents a comparison between individual model performances, our upscaled SMILE model, and a MergeKit-MoE [42] baseline across multiple benchmark tasks. Notably, individual expert models demonstrate strengths in specific benchmarks, such as M_1 excelling in the GSM8K benchmark and M_3 in the ARC Challenge. This indicates that each expert brings specialized knowledge, which when combined, enhances the overall performance in a diverse set of tasks. Further details are provided in Appendix G.

VI. RELATED WORK

1) *Mixture of Experts*: The concept of Mixture of Experts (MoE) is first introduced by [44], involving training multiple specialized models. This concept has gained significant attention in recent years [45], [46], with much of the innovation focusing on routing mechanisms and expert design. Much innovation revolves around the design of more efficient routers. For example, the Switch Transformer [47] selects only the top expert for each token, simplifying the process and improving scalability. Similarly, [48] use a linear assignment to optimize token-expert affinities, ensuring an equal spread of tokens among experts. For detailed reviews on MoE, see [49] and for MoE in the context of model merging, refer to [50].

1) *Deep Model Fusion*: Model connectivity reveals that different model solutions can be linked by low-loss path in the parameter space [51], [52], [53], [54], [55], [56], [57], [58], [59], facilitating model fusion by weight interpolation [11], [17], [33], [34], [35], [60], [61], [62]. However, this strategy also poses challenges, particularly when merging models with diverse structures. Alignment helps reduce model disparities by matching and interpolating components [57], [63]. Methods involve matching activations or weights [12], [16], [36], [64], using channel-wise graph matching [65], or applying permutation invariance [15]. Another line of research is model mixing, which combines models through gating mechanisms or depth concatenation [14], [39], [66], [67], allowing for more flexible and adaptive fusion strategies.

VII. CONCLUSION, LIMITATIONS, AND FUTURE WORK

In this paper, we introduced the Sparse Mixture of Low-Rank Experts (SMILE) model as an innovative approach to model fusion. Our method is zero-shot, thus eliminating the need for additional training data or processes, making it highly practical and efficient. Although the Mixture of Experts (MoE) method aims to be efficient through sparse activation, it still incurs additional computational overhead, particularly as the number of tasks or experts increases. Additionally, our method shares the inherent limitation of weight interpolation-based fusion approaches when

applied to models that have substantially diverged from their shared foundation.

Understanding which subspaces contribute most to task-specific performance could enable the development of more targeted and efficient fine-tuning strategies. This could involve focusing on updating specific parts of the model while leaving others unchanged. Additionally, this approach might be applied to other areas, like multi-modal large language models, where different types of data (modalities) are treated as separate experts. Furthermore, exploring how SMILE can manage multi-objective optimization by adjusting the importance of different routing weights could provide valuable insights. Another avenue for development is creating methods to dynamically adjust the number of experts K per token based on the input. This could enhance efficiency without compromising performance.

ACKNOWLEDGMENT

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Anke Tang (Graduate Student Member, IEEE) received the bachelor's degree in 2020 from the School of Physics and Technology, Wuhan University, where he is currently working toward the PhD degree with the School of Computer Science, under the supervision of prof. Yong Luo. He has authored or coauthored papers in proceedings at leading conferences, such as IJCAI, ICLR, and ICML. His research interests include machine learning, transfer learning, and multi-task learning. He was a reviewer of several top-tier conferences and journals.



Li Shen (Member, IEEE) received the bachelor's and PhD degrees from the School of Mathematics, South China University of Technology. He was a research scientist with JD Explore Academy, Beijing, and a senior researcher with Tencent AI Lab, Shenzhen. He is currently an associate professor with Sun Yat-sen University. He has authored or coauthored more than 100 papers in peer-reviewed top-tier journal papers such as *JMLR*, *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *IJCV*, *IEEE Transactions on Signal Processing*, *IEEE Transactions on Image Processing*, *IEEE Transactions on Knowledge and Data Engineering*, and conference papers such as *ICML*, *NeurIPS*, *ICLR*, *CVPR*, and *ICCV*. His research interests include theory and algorithms for nonsmooth convex and non-convex optimization, and their applications in trustworthy artificial intelligence, deep learning, and reinforcement learning. He was the senior program committee for AAAI and area chairs for *ICML*, *ICLR*, *ICPR*, and *ACML*.



Yong Luo received the BE degree in computer science from Northwestern Polytechnical University, Xi'an, China, and the DSc degree from the School of Electronics Engineering and Computer Science, Peking University, Beijing, China. He is currently a professor with the School of Computer Science, Wuhan University, China. He has authored or co-authored more than 80 papers in top journals and prestigious conferences including *Nature Communications*, *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *IEEE Transactions on Neural Networks and Learning Systems*, *IEEE Transactions on Image Processing*, *IEEE Transactions on Knowledge and Data Engineering*, *IEEE Transactions on Multimedia*, *CVPR*, *ICCV*, *WWW*, *IJCAI*, and *AAAI*. His research interests include machine learning and data mining with applications to visual information understanding and analysis. He is on the editorial board of *IEEE Transactions on Multimedia*. He was the recipient of the IEEE Globecom 2016 Best Paper Award, and was nominated as the IJCAI 2017 Distinguished Best Paper Award. He is also a co-recipient of the IEEE TMM 2023, IEEE ICME 2019, and IEEE VCIP 2019 Best Paper Awards.



Shuai Xie received the bachelor's degree from the School of Computer Science and Engineering, Central South University, and the master's degree from the College of Computer Science and Technology, Zhejiang University. He is currently a research engineer with JD Explore Academy, Beijing. His research interests include cost and effective large language model inference technology, including quantization, sparsity, speculative decoding, and their best practice in the industrial applications.



Paper Award of the IEEE Globecom 2013. He was an associate editor for *IEEE Transactions on Multimedia* and *Ad Hoc Networks*, and a TPC member of Infocom, ACM MM, AAAI, and IJCAI.

Han Hu (Member, IEEE) received the BE and PhD degrees from the University of Science and Technology of China, in 2007 and 2012, respectively. He is currently a professor with the School of Information and Electronics, Beijing Institute of Technology, China. His research interests include multimedia networking, edge intelligence, and space-air-ground integrated network. He was the recipient of several academic awards, including the Best Paper Award of the IEEE TCSVT 2019, the Best Paper Award of the IEEE Multimedia Magazine 2015, and the Best



recognition, hyperspectral image processing, and signal processing.

Bo Du received the PhD degree from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, in 2010. He is currently a professor with the School of Computer, Wuhan University. He has authored or coauthored more than 100 scientific papers, such as *IEEE Transactions on Geoscience and Remote Sensing*, *IEEE Transactions on Neural Networks and Learning Systems*, *IEEE Transactions on Image Processing*, *IEEE Transactions on Cybernetics*, AAAI, and IJCAI. His research interests include pattern



recognition, image processing, and remote sensing. He is a topical editor of *IEEE Transactions on Geoscience and Remote Sensing*, an associate editor for *Pattern Recognition*, and the section editor-in-chief for *Remote Sensing*.

Lefei Zhang (Senior Member, IEEE) received the BS and PhD degrees from Wuhan University, Wuhan, China, in 2008 and 2013, respectively. He was a Big Data Institute visitor with the Department of Statistical Science, University College London, U.K., and a Hong Kong scholar with the Department of Computing, The Hong Kong Polytechnic University, Hong Kong, SAR, China. He is currently a professor with the School of Computer Science, Wuhan University, and with Hubei LuoJia Laboratory, Wuhan, China. His current research interests include pattern



Scholar. He was the recipient of the 2015 and 2020 Australian Eureka Prize, the 2018 IEEE ICDM Research Contributions Award, and the 2021 IEEE Computer Society McCluskey Technical Achievement Award. He is a Fellow of the Australian Academy of Science, AAAS, and ACM.

Dacheng Tao (Fellow, IEEE) is currently a distinguished university professor with the College of Computing & Data Science, Nanyang Technological University. He mainly applies statistics and mathematics to artificial intelligence and data science, and his research is detailed in one monograph and more than 200 publications in prestigious journals and proceedings at leading conferences, with best paper awards, best student paper awards, and test-of-time awards. His publications have been cited more than 112 K times and he has an h-index 160+ in Google